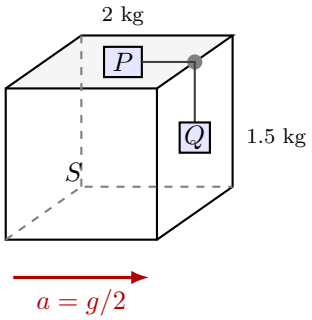
Level 4 Sheet

Time to be Taken: 90 Min Max

1. Two blocks (P and Q) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionless pulley which is fixed on the edge of a cube (S), as shown in the figure below. Block P is positioned on the top surface which has no friction and block Q is in contact with side-surface, having coefficient friction μ. The cube (S) moves towards the right with acceleration of $\frac{g}{2}$, where g is gravitational acceleration. During this movement the block P and Q remain stationary. The value of μ is _____. (take $g = 10$ m/s^2)

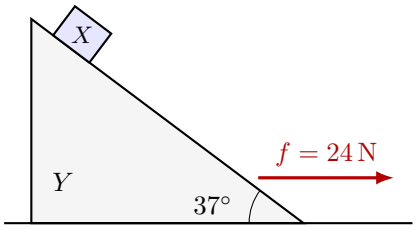


(A) 0.33 (B) 0.67

(C) 1 (D) 0.5

(Pseudo Concept Involved)

2. A wedge Y with mass of 10 kg and all frictionless surfaces and the inclined surface making 37° with horizontal. A block X with mass 2 kg is placed at the highest point of the wedge as shown in figure is at rest. At $t = 0$ wedge (Y) is pulled toward right with constant force (f) of 24 N. Taking the block X at rest at $t = 0$, the time taken by it to slide down 8.8 m on the slope, while Y is on the move, is _____ s. (take $\tan(37°) = 3/4$ and $g = 10$ m/s^2)

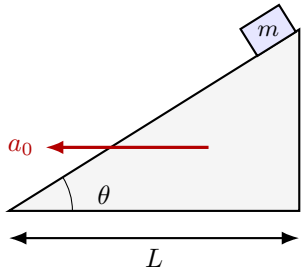


(A) 2 (B) 4

(C) $\sqrt{2}$ (D) $2\sqrt{2}$

(Pseudo Concept Involved)

3. A small block of mass m slides down from the top of a frictionless inclined surface, while the inclined plane is moving towards left with constant acceleration a_0. The angle between the inclined plane and ground is θ and its base length is L. Assuming that initially the small block is at the top of the inclined plane, the time it takes to reach the lowest point of the inclined plane is _____.



(A) $\sqrt{\dfrac{2L}{g\sin\theta - a_0\cos\theta}}$

(B) $\sqrt{\dfrac{4L}{g\cos^2\theta - a_0\sin\theta\cos\theta}}$

(C) $\sqrt{\dfrac{2L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$

(D) $\sqrt{\dfrac{4L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$

(Pseudo Concept Involved)

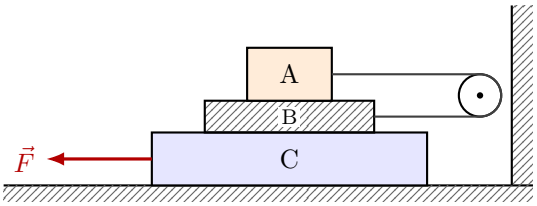
4. A block takes t time to slide down a plane inclined at 45° to the horizontal. If the surface is made smooth (frictionless), the block takes time $\dfrac{t}{2}$ to slide down the plane. The coefficient of friction between the block and the inclined plane is $\left(\dfrac{\alpha}{100}\right)$. The value of α is _____.

(Very Important, practice all variations)

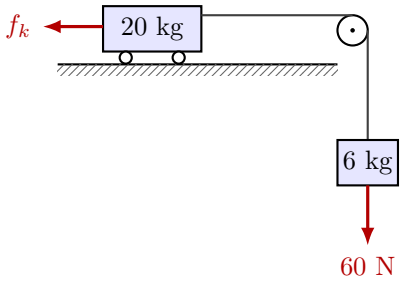
5. The time taken by a block of mass m to slide down from the highest point to the lowest point on a rough inclined plane is 50% more compared to the time taken by the same block on identical inclined smooth plane. Both inclined planes are at 45° with the horizontal. The coefficient of kinetic friction between the rough inclined surface and block is _____.

(Similar to the above question)

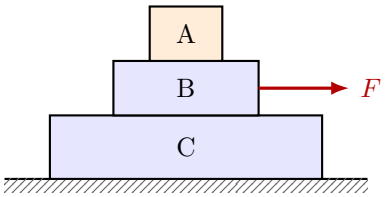
6. In the given figure the blocks A, B and C weigh 4 kg, 6 kg and 8 kg respectively. The co-efficient of sliding friction between any two surfaces is 0.5. The force $\vec{F}$ required to slide the block C with constant speed is _____ N. (Use $g = 10$ m/s^2)



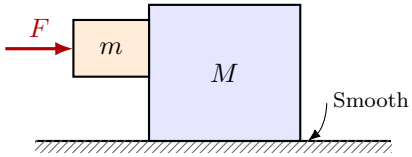
7. Consider a block and trolley system as shown in figure. If the coefficient of kinetic friction between the trolley and the surface is 0.04, the acceleration of the system in $\mathrm{m\,s^{-2}}$ is: (Consider that the string is massless and unstretchable and the pulley is also massless and frictionless) :



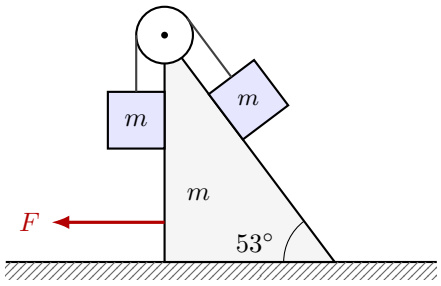
8. In the figure shown, the coefficient of static friction between C and ground is 0.5, coefficient of static friction between A and B is 0.25, coefficient of static friction between B and C is zero. Find the minimum value of force 'F' (in N), to cause sliding between A and B. Masses of A, B and C are respectively 2 kg, 4 kg and 5 kg.



9. The two blocks, $m = 10\,\mathrm{kg}$ and $M = 50\,\mathrm{kg}$ are free to move as shown. The coefficient of static friction between the blocks is 0.5 and there is no friction between M and the ground. A minimum horizontal force F is applied to hold m against M that is equal to $N \times 60$ newton then find N :

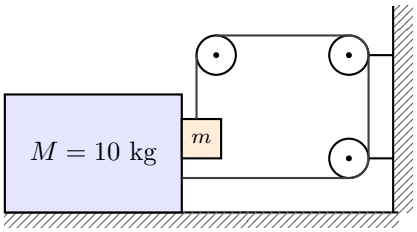


10. A wedge of mass m is kept on a smooth horizontal surface. Two blocks, each of mass m are also attached with the wedge as shown. Separation between block and vertical wall of wedge is negligible. If all the surfaces are smooth, how much horizontal force F should be applied to the wedge, so that the blocks does not slide with respect to wedge ?



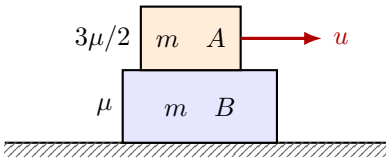
(1) $\frac{5mg}{3}$ (2) $\frac{3mg}{5}$ (3) mg (4) $3\,mg$

11. The maximum value of m(in kg) so that the arrangement shown in the figure is in equilibrium is (friction coefficient between block and wedge is 0.1 and between wedge and ground is 0.4)



(1) 2 (2) 2.5 (3) 3 (4) 3.5

12. In the figure shown a block A of mass m is kept on block B of mass m. A is given a velocity towards right as shown. The coefficient of friction between A and B is $\frac{3\mu}{2}$ and that between B and ground is μ.



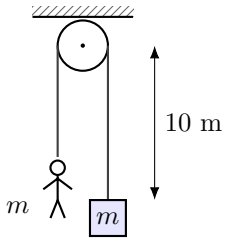
(1) B will accelerate towards right with acceleration $\frac{\mu g}{2}$.

(2) B will accelerate towards left with acceleration $\frac{\mu g}{2}$.

(3) B will not accelerate at all.

(4) B will move towards right with acceleration μg.

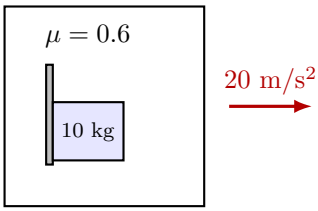
13. A boy and a block, both of the same mass 10 kg, are suspended at the same horizontal level, from each end of a light string that moves over a frictionless pulley as shown. The boy starts moving upwards with an acceleration 2.5 m/s^2 relative to the rope. Then the tension in the string will be : ($g = 10$ m/s^2)



(1) 112.5 N (2) 125 N (3) 200 N (4) 100 N

(Revise Monkey climbing cases from notes, make sure to note which acceleration is given whether w.r.t. rope or ground)

14. Car is accelerating with acceleration $= 20\ \text{m/s}^2$. A box of mass $m = 10\,\text{kg}$ that is placed inside the car, it is put in contact with the vertical wall of car as shown. The friction coefficient between the box and the wall is $\mu = 0.6$.



(A) The acceleration of the box will be $20\ \text{m/s}^2$

(B) The friction force acting on the box will be $100\,\text{N}$

(C) The contact force between the vertical wall and the box will be $100\sqrt{5}\,\text{N}$

(D) The net contact force between the vertical wall and the box is only of electromagnetic in nature.

Keep Going!

"Like friction, the struggle you face today is exactly what gives you the grip to climb tomorrow. Keep pushing — your breakthrough is closer than you think."